



RV College of Engineering
Mysuru Road, RV Vidyanikethan Post,
Bengaluru - 560 059, India

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R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

I / II Semester B.E Regular/Supplementary Examinations July/August-2025.

Common to all

Course : Principles Of Electronics Engineering-EC113ATC/EC123ATC

Time : 3 Hours

Maximum Marks : 100

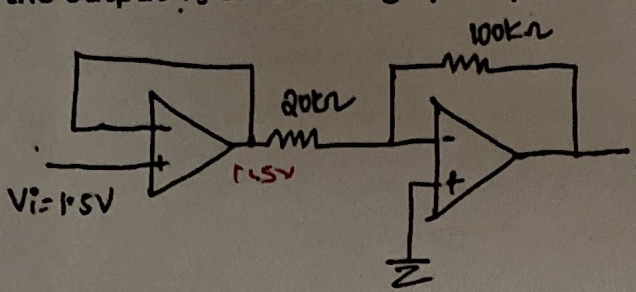
Instructions to the students

1. Answer all the questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and four, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	A photodiode generates current when exposed to _____.	01	1	1
1.2	In CE amplifier, the terminal common to both input and output is the _____.	01	3	2
1.3	The overall gain of cascaded stages in dB is the _____ of individual stage gains in dB.	01	3	3
1.4	Cascading amplifiers increases the overall _____.	01	3	2
1.5	In a RC phase shift oscillator circuit using an ideal voltage amplifier with $C=0.01\mu\text{F}$ and $R=2\text{K}\Omega$. The frequency of oscillation is _____.	02	2	2
1.6	With negative feedback, the bandwidth of an amplifier _____ by a factor $(1+A\beta)$	01	1	1
1.7	Why negative feedback is used in op-amp circuits.	01	3	2
1.8	Write the 2's complement of $(11001)_2$.	01	1	2
1.9	Convert $(B6)_{16}$ to binary.	01	1	2
1.10	Write the truth table for a 2-input NAND gate.	02	2	2
1.11	Simplify the expression using K-maps: $F(A,B,C)=\Sigma(0,1,2,3)$	02	1	4
1.12	The amplitude modulation spectrum consists of _____ sideband frequencies and also a carrier frequency	01	1	1
1.13	An AM signal has a carrier frequency of 1 MHz and the highest modulating frequency of 5 kHz. Calculate the bandwidth of the AM signal	02	2	2
1.14	The resistance of a platinum RTD at 0°C is _____	01	1	1
1.15	Mention any one application of Piezoelectric transducers	01	1	1
1.16	_____ sensors is used to detect ambient light.	01	1	1

Part B

Question No	Question	M	CO	BT
2a	Explain the design of a Zener regulator with varying V_{in} and I_L , and derive the conditions for selecting R so that I_Z stays between $I_{Z(min)}$ and $I_{Z(max)}$.	08	2	2
2b	With a neat common emitter circuit diagram, explain about different current components in a NPN transistor and also find β_{dc} in terms of α_{dc} and viceversa.	08	3	3
3a	Find the output V_o of following op-amp circuit.	06	2	3
 $V_i = 1.5V$ $V_o = -\left(\frac{R_f}{R_i}\right) V_i$ $= -\left(\frac{100k}{20k}\right) V_i$ $= -5 V_i$ $= -7.5V$				
3b	A voltage amplifier needs 2 mV input to give a 10V output. When negative feedback is provided to this amplifier, it needs 200mV to deliver the same output. Determine the open loop gain, closed loop gain, feedback factor and amount of feedback in dB for this amplifier.	04	3	3
3c	Explain the principle of operation of Wein Bridge Oscillator with relevant equations to find the resonant frequency.	06	3	2
OR				
4a	An amplifier has a gain of 50 dB. The bandwidth of 250KHz, distortion of 12%, an input impedance of 30K Ω , and an output impedance of 2K Ω . If the voltage series negative feedback of 2.9% is given to this amplifier, calculate the gain, input impedance, output impedance, bandwidth, and distortion of the amplifier with negative feedback.	08	3	4
4b	List at least Eight important characteristics of an ideal op-amp and indicate their typical values for a general purpose commercial op-amp. Give the Pin configuration of OPAMP-741	08	3	2
5a	Simplify the expressions: i. $F_1 = (A + B)(A + C)$ ii. $F_2 = A + AB + AB'$	04	1	2
5b	Write the truth table for SUM and CARRYOUT of a full adder. From the truth table, obtain the logic expressions for the same and then realize the full adder using 2 half adders.	06	2	3
5c	Minimize the following boolean function using four variables K-Map and implement using NOR gate only	06	3	5
6a	Simplify the Boolean expressions $F_1 = A'B + AB + AB'$ and $F_2 = A + AB + AB'C$	04	2	2

AB
 A ⊕ B
 V_C = 25
 m = 0.3
 f_m = 500
 4C = 5 MHz

f_C = 4.99 MHz
 f_m = 499 Hz
 V_C = 20 V
 m = 0.6
 B.W. = 99.4 kHz

Design a half adder circuit using Basic logic gates with necessary equations and truth table also implement the half adder using NOR gates only. 06 2 4

Simplify using K-map and implement using NAND gates: 06 3 4

$F(A,B,C,D) = \sum m(0,2,3,6,8,10,11,14)$ $\overline{C}D + C\overline{B} + B\overline{D}$

Draw the block diagram of digital communication system and explain each block 08 1 1

Calculate fm, fc, Vc, m, bandwidth and Pt for the amplitude modulated wave given by 08 3 3

$S(t) = 25[1 + 0.3\sin 3140t]\sin 31.4 \times 10^8 t$

OR

List 4 differences of the following:

8a i. Harvard and Von-Neumann architecture. 08 2 2

ii. Microprocessor and microcontroller

Calculate fm, fc, Vc, m, bandwidth and Pt for the amplitude modulated wave given by $S(t) = 20[1 + 0.2\sin 3140t]\sin 31.4 \times 10^6 t$ with Pc=4kW 08 3 3

9a Compare and contrast active and passive transducers with respect to their working principle, operation characteristic and applications. 06 1 1

9b Discuss how proximity sensors are implemented in robotics and highlight their benefits. 06 1 1

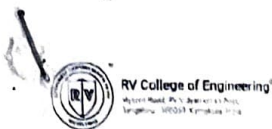
9c Illustrate the resistance-temperature characteristic of an RTD sensor with a suitable graph. 04 1 1

OR

10a Describe the use of touch sensor technology in mobile phones, highlighting its working principle. 04 1 1

10b Describe a real-life example where a transducer is used in automation. 06 1 1

10c Discuss the selection parameters for a transducer used in high-temperature furnace applications. 06 1 1

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R V College of EngineeringR V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059**II Semester B.E Regular/Supplementary Examinations July-2025.**
Common to all
Course : Principles Of Electronics Engineering-EC113AT/EC123AT**Time : 3 Hours****Maximum Marks : 100****Instructions to the students**

1. Answer all the questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and four, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
	A photodiode generates current when exposed to ____.			
	Solution:			
1.1	Light	01	1	1
	Solution Scheme:			
	In CE amplifier, the terminal common to both input and output is the ____.			
	Solution:			
1.2	emitter	01	3	2
	Solution Scheme:			
	The overall gain of cascaded stages in dB is the ____ of individual stage gains in dB.			
	Solution:			
1.3	sum	01	3	3
	Solution Scheme:			
	Cascading amplifiers increases the overall ____.			
	Solution:			
1.4	gain	01	3	2
	Solution Scheme:			
	In a RC phase shift oscillator circuit using an ideal voltage amplifier with $C=0.01\mu F$ and $R=2K\Omega$. The frequency of oscillation is ____.			
	Solution:			
1.5	3.25 kHz	02	2	2
	Solution Scheme:			
1.6	With negative feedback, the bandwidth of an amplifier ____ by a factor $(1+A\beta)$	01	1	1
	Solution:			
	Increases			

Solution Scheme:

Why negative feedback is used in op-amp circuits.

Solution:

- 1.7 **Negative feedback makes op-amps practical** by improving **stability, accuracy, linearity, and bandwidth** while reducing noise and distortion.

01 3 2

Solution Scheme:

Write the 2's complement of $(11001)_2$.

Solution:

- 1.8 Answer: $(00111)_2$

01 1 2

Solution Scheme:

1's complement of $11001 = 00110$
Add 1: $00110 + 1 = 00111$
Convert $(B6)_{16}$ to binary.

Solution:

- 1.9 $(10110110)_2$

01 1 2

Solution Scheme:

$B = 1011, 6 = 0110 \rightarrow (B6)_{16} = (10110110)_2$
Write the truth table for a 2-input NAND gate.

Solution:

- 1.10
- | A | B | A NAND B |
|---|---|----------|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

02 2 2

Solution Scheme:

A	B	A NAND B
0	0	1
0	1	1
1	0	1
1	1	0

Simplify the expression using K-maps: $F(A,B,C) = \Sigma(0,1,2,3)$

Solution:

A'

- 1.11

Solution Scheme:

02 1 4

A \ BC	00	01	11	10
0	1	1	1	1
1	0	0	0	0

- 1.12 The amplitude modulation spectrum consists of _____ sideband frequencies and also a carrier frequency

01 1 1

Solution:

Solution Scheme:

1. 2 sidebands

An AM signal has a carrier frequency of 1 MHz and the highest modulating frequency of 5 kHz. Calculate the bandwidth of the AM signal

Solution:

- 1.13 1. 10 KHz

02 2 2

Solution Scheme:

The resistance of a platinum RTD at 0°C is _____

Solution:

- 1.14 100 ohm

01 1 1

Solution Scheme:

Mention any one application of Piezoelectric transducers

Solution:

- 1.15 Pressure and vibration measurement.

01 1 1

Solution Scheme:

_____ sensors is used to detect ambient light.

Solution:

- 1.16 Light Dependent Resistor (LDR)

01 1 1

Solution Scheme:

Part B

**Question
No**

- 2a Explain the design of a Zener regulator with varying V_{in} and I_L , and derive the conditions for selecting R so that I_Z stays between

Solution:

Figure: 2M and Explanation:6M



In the above figure both input voltage V_{in} and load R_L are varying. When we need to design a zener regulator, the parameters like considered

Here V_{in} varies between $V_{in(min)}$ to $V_{in(max)}$ and the load current I_L varies from $I_L(min)$ to $I_L(max)$. The calculation of R should should operate between $I_Z(min)$ and $I_Z(max)$. The current through zener must be more than $I_Z(min)$, where $I_Z(min)$ is the minimum to operate in the breakdown region.

$$[(V_{in(min)} - V_Z)/R] - I_L(max) > I_Z(min)$$

Maximum zener current flows when $V_{in} = V_{in(max)}$ and $I_L = I_L(min)$. the current through zener must be less than $I_Z(max)$, where $I_Z(max)$ is the allowable zener current for safe operation.

$$[(V_{in(max)} - V_z)/R] - I_{L(min)} < I_{z(max)}$$

Where $I_{z(max)} = P_D / V_z$

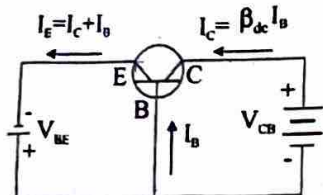
P_D is the maximum allowable power dissipation in zone.

Solution Scheme:

With a neat common emitter circuit diagram, explain about different current components in a NPN transistor and also find β_{dc} in viceversa .

Solution:

Solution: Circuit diagram: 1M, Different current components: 2M, β_{dc} in terms of α_{dc} and viceversa : 5M



for npn transistor, I_E & I_C are assumed to flow into device & I_B is assumed to flow out of the device as shown in above figure.

$$I_E = I_C + I_B \quad (1)$$

$$\text{But } I_E = \alpha_{dc} I_C$$

$$\therefore I_C = \alpha_{dc} (I_C + I_B)$$

2b

$$\therefore I_C = \frac{\alpha_{dc}}{1 - \alpha_{dc}} I_B$$

$$I_C = \beta_{dc} I_B$$

$$\beta_{dc} = \frac{I_C}{I_B}$$

$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}}$$

Example 1:

$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}}$$

$$\text{or } \beta_{dc} (1 - \alpha_{dc}) = \alpha_{dc}$$

$$\beta_{dc} - \beta_{dc} \alpha_{dc} = \alpha_{dc}$$

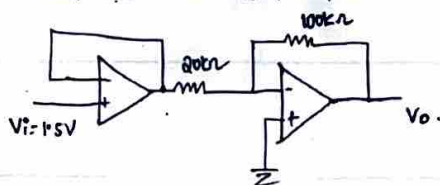
$$\beta_{dc} = \alpha_{dc} + \beta_{dc} \alpha_{dc}$$

$$\beta_{dc} = (1 + \beta_{dc}) \alpha_{dc}$$

$$\therefore \alpha_{dc} = \frac{\beta_{dc}}{1 + \beta_{dc}}$$

Solution Scheme:

3a Find the output V_o of following op-amp circuit.



Solution:

- Input voltage: $V_i = 1.5 \text{ V}$
- First op-amp: Voltage follower $\rightarrow$ output is still 1.5 V
- Second stage: Inverting amplifier
 - Input resistor: $R_1 = 20 \Omega$
 - Feedback resistor: $R_f = 100 \text{ k}\Omega = 100,000 \Omega$

$$V_o = - \left(\frac{R_f}{R_i} \right) \cdot V_{in}$$

Substitute values:

$$V_o = - \left(\frac{100,000}{20} \right) \cdot 1.5 = -5000 \cdot 1.5 = -7500 \text{ V (theoretical)}$$

Solution Scheme:

Stage 1 op= 3M

Final Vo=3M

A voltage amplifier needs 2 mV input to give a 10V output. When negative feedback is provided to this amplifier, it needs 200mV output. Determine the open loop gain, closed loop gain, feedback factor and amount of feedback in dB for this amplifier.

Solution:

3b

Solution Scheme:

A=5000=73dB---1M

Af=50=33.9dB-----1M

B=0.0198-----2M

Explain the principle of operation of Wein Bridge Oscillator with relevant equations to find the resonate frequency.

Solution:

3c

Solution Scheme:

circuit-3M

Explanation-3M

OR

An amplifier has a gain of 50 dB. The bandwidth of 250KHz, distortion of 12%, an input impedance of 30KΩ, and an output impedance series negative feedback of 2.9% is given to this amplifier, calculate the gain, input impedance, output impedance, bandwidth the amplifier with negative feedback.

Solution:

4a

Solution Scheme:

Gain

307.35 (or 49.75 dB)--2M

Bandwidth

257.25 kHz---2M

Distortion

11.66%---1M

Input Impedance

30.87 kΩ---1M

Output Impedance

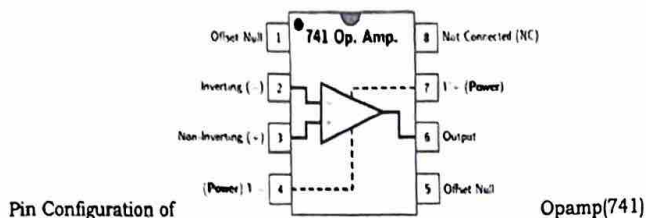
1.944 kΩ---2M

4b

List at least Eight important characteristics of an ideal op-amp and indicate their typical values for a general purpose commercial configuration of OPAMP-741

Solution:

Parameter	Ideal	Typical or Practical Value
Voltage Gain $ A_v $	∞	2×10^5
Output Impedance	0	75Ω
Input Impedance	∞	2MΩ
Input Offset	0	2mV
CMRR	∞	90dB
Slew Rate	∞	0.5V/μs
Bandwidth	∞	1MHz
PSRR	0	30μV/V
Input Bias Current	0	80nA



Solution Scheme:

characteristics-4M

Pin diagram-4M

- Simplify the expressions $F_1 = (A + B)(A + C)$ and
- $F_2 = A + AB + AB'$

Solution:

- 5a $F_1 = A + BC$
 $F_2 = A$

Solution Scheme:

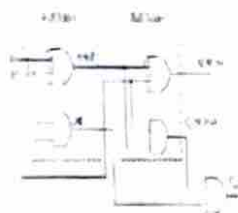
- 5b Write the truth table for SUM and CARRYOUT of a full adder. From the truth table, obtain the logic expressions for the same and adder using 2 half adders.

Solution:

$$\begin{aligned}
 \text{Sum} &= \overline{A} \overline{B} C_{in} + \overline{A} B \overline{C}_{in} + A \overline{B} \overline{C}_{in} + ABC_{in} \\
 &= \overline{A} [\overline{B} C_{in} + B \overline{C}_{in}] + A [\overline{B} \overline{C}_{in} + B C_{in}] \\
 &= \overline{A} [B \oplus C_{in}] + A [\overline{B} \oplus C_{in}] \\
 &= A \oplus B \oplus C_{in} \\
 \text{Carry} &= \overline{A} B C_{in} + A \overline{B} C_{in} + AB \overline{C}_{in} + ABC_{in} \\
 &= \overline{A} B C_{in} + A \overline{B} C_{in} + AB (\overline{C}_{in} + C_{in}) \\
 &= \overline{A} B C_{in} + A \overline{B} C_{in} + AB \\
 &= \overline{A} B C_{in} + A (\overline{B} C_{in} + B) \\
 &= \overline{A} B C_{in} + AB + AC_{in} \\
 &= B (\overline{A} C_{in} + A) + AC_{in} \\
 &= B (A + C_{in}) + AC_{in} \\
 &= AB + BC_{in} + AC_{in}
 \end{aligned}$$

Solution Scheme:

Input			Output	
A	B	C_{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



5c Minimize the following boolean function using four variables K-Map and implement using NOR gate only

$$F(P, Q, R, S) = \sum m(0, 1, 2, 4, 5, 6, 8, 10, 11, 12, 14, 15)$$

Solution:

Solution Scheme:

RS		00	01	11	10
PQ	00	1	1		1
	01	1	1		1
	11	1		1	1
	10	1		1	1

$$y = \bar{P}\bar{R} + RP +$$

OR

Simplify the Boolean expressions $F_1 = A'B + AB + AB'$ and $F_2 = A + AB + AB'C$

Solution:

6a $F_1 = A+B$ and $F_2 = A$

Solution Scheme:

Design a half adder circuit using Basic logic gates with necessary equations and truth table also implement the half adder using I

Solution:

Truth Table for Half Adder

Input		Output	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

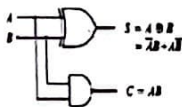
Logic Expression:

$$\text{Sum} = \bar{A}B + A\bar{B} = A \oplus B$$

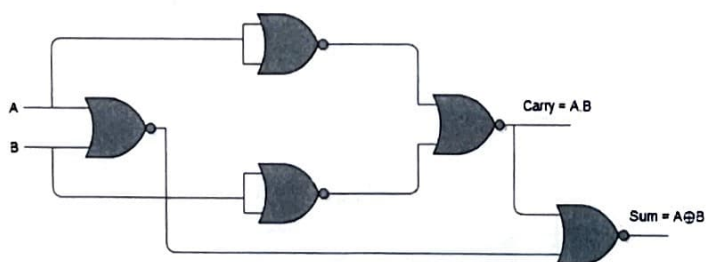
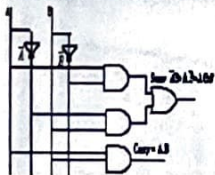
$$\text{Carry} = A.B$$

Solution Scheme:

6b



The circuit for Half Adder using Basic Gates is as follows:



6c Simplify using K-map and implement using NAND gates:

$$F(A,B,C,D) = \sum m(0,2,3,6,8,10,11,14)$$

Solution:

$$CD' + CB' + B'D'$$

Solution Scheme:

AB \ CD				
	00	01	11	10
00	1	0	1	1
01	4	5	7	1
11	12	13	15	1
10	1	8	9	11

$$Y = C\bar{D} + C\bar{B} + \bar{B}$$

Draw the block diagram of digital communication system and explain each block

Solution:



7a

Solution Scheme:

Calculate f_m , f_c , V_c , m , bandwidth and P_t for the amplitude modulated wave given by

$$S(t) = 25[1 + 0.3\sin(3140t)]\sin(31.4 \times 10^8 t)$$

Solution:

7b

Solution Scheme:

- $V_c = 25$ volts (carrier amplitude)
- $m = 0.3$ (modulation index)
- $2\pi f_m = 3140 \Rightarrow f_m = 500$ Hz
- $2\pi f_c = 31.4 \times 10^8 \Rightarrow f_c = 5$ MHz

OR

8a List 4 differences of the following:

- i. Harvard and Von-Neumann architecture.
- ii. Microprocessor and microcontroller

Solution:

Solution Scheme:

i.

Feature	Von Neumann Architecture	Harvard Architecture
Memory	Shared for code and data	Separate for code and data
Buses	Single bus for both	Separate buses
Execution Speed	Slower (due to bus conflict)	Faster (parallel access)
Design Complexity	Simpler	More complex
Cost	Lower	Slightly higher
Usage	General-purpose CPUs	Embedded systems, microcontrollers
Feature	Microprocessor	8. Microcontroller
Components	Only CPU	CPU, RAM, ROM, I/O ports (all-in-one)
Memory	External	Internal
Application	General-purpose computing	Embedded/control systems
Cost	Higher (needs extra components)	Lower (everything in one chip)
Speed	Higher (optimized for computation)	Moderate (optimized for control)
Power Consumption	High	Low

ii.

S.No. Microprocessor

1. Contains **only CPU** (Central Processing Unit).
2. Requires **external memory** and peripherals to function.
3. Used in **general-purpose** computing devices.
4. **High power consumption** due to external interfacing.
5. **Expensive** system due to extra components.
6. Focused on **processing speed and performance**.

Microcontroller

Contains **CPU, RAM, ROM, I/O ports, and timers** on a single chip. Has **built-in memory** (RAM, ROM/Flash) and I/O ports. Used in **embedded systems** and control applications. **Low power consumption** – suitable for battery-operated systems. **Cost-effective** due to integration of components. Focused on **real-time control tasks** and reliability.

S.No. Microprocessor

7. Typically found in **PCs, laptops, servers**.
 8. Needs **more space** on the PCB (printed circuit board).

Microcontroller

Found in **microwaves, washing machines, automobiles**.
Compact design – ideal for space-constrained applications.

Calculate f_m , f_c , V_c , m , bandwidth and P_t for the amplitude modulated wave given by $S(t) = 20[1 + 0.2\sin 3140t]\sin 31.4 \times 10^6 t$ with P_c

Solution:

Solution Scheme:

8b

- $f_c = 4.99 \text{ MHz}$
- $f_m = 499.74 \text{ Hz}$
- $V_c = 20 \text{ Volts}$
- $\mu = 0.6$
- Bandwidth = 999.48 Hz

Compare and contrast active and passive transducers with respect to their working principle, operation characteristic and applications.

Solution:

Solution Scheme:

9a

Aspect	Active Transducers	Passive Transducers
Working Principle	Generate electrical output directly from a physical quantity (self-generating)	Require external power to convert a physical quantity into an electrical signal
Energy Source	Do not need an external power source	Need an external excitation or power source
Operation Characteristic	Operate by converting energy from non-electrical to electrical (e.g., heat, pressure to voltage)	Operate by varying electrical properties (e.g., resistance, capacitance, inductance)
Examples	Thermocouple, Piezoelectric sensor, Solar cell	RTD, Strain gauge, LVDT, Photodiode
Output Signal Type	Typically voltage or current	Typically change in resistance, capacitance, inductance
Applications	Used where power supply is difficult (remote sensing, shock/vibration)	Used in precision measurement systems (displacement)
Sensitivity	Generally less sensitive to small inputs	Generally more sensitive with better resolution

Discuss how proximity sensors are implemented in robotics and highlight their benefits.

Solution:

A **proximity sensor** is a type of transducer that detects the presence or absence of an object **without physical contact**. It is crucial for **navigation, object detection, automation, and safety**.

Implementation in Robotics

a. Obstacle Detection and Avoidance

- Robots use ultrasonic or IR proximity sensors to **sense obstacles** in their path.
- If an object is detected within a threshold range, the robot **slows down, stops, or changes direction**.

b. Gripper Automation

- Proximity sensors help detect whether an object is in place for **picking or placing**.
- Inductive sensors in industrial robot grippers can confirm the **presence of a metallic component**.

9b

c. Navigation and Mapping

- In mobile robots and autonomous vehicles, **LiDAR and ultrasonic sensors** are used to:
 - o Map surroundings (SLAM: Simultaneous Localization and Mapping)
 - o Avoid collisions
 - o Navigate dynamic environments

d. Safety and Human-Robot Interaction

- Capacitive proximity sensors are used to detect human presence near collaborative robots (cobots).
- Enables **safe stopping or speed reduction** when humans are nearby.

Solution Scheme:

Illustrate the resistance-temperature characteristic of an RTD sensor with a suitable graph.

Solution:

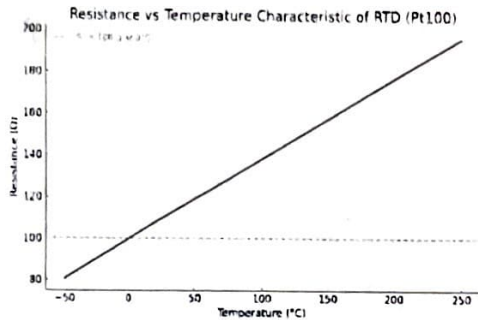
The graph above illustrates the **Resistance-Temperature characteristic** of an RTD (specifically a Pt100 sensor). As seen:

The resistance increases **linearly** with temperature.

At **0°C**, the resistance is **100 Ω**, which is the standard for a Pt100 RTD.

The linear rise is due to the **positive temperature coefficient** of platinum.

This linearity makes RTDs highly suitable for accurate temperature measurement in industrial applications.



illustrates the **Resistance-Temperature characteristic** of an RTD (specifically a Pt100

Solution Scheme:

OR

Describe the use of touch sensor technology in mobile phones, highlighting its working principle.

Solution:

Touch sensors in mobile phones are a core part of the **touchscreen interface**, allowing users to interact with their devices through swiping, and pinching. Most modern smartphones use **capacitive touch technology**, which is reliable, durable, and supports multi-touch gestures.

How It Works:

1. **Conductive Layer Grid:**

- o The screen contains a grid of **transparent conductive materials** (like indium tin oxide) forming **rows and columns**.
- o This grid acts as a matrix of **capacitors** at each intersection.

2. **Electric Field:**

- o A small voltage is applied across the grid, creating a **uniform electrostatic field**.

3. **Touch Detection:**

- o When a finger touches the screen, it **disturbs the electrostatic field** at the point of contact because the human body

4. **Capacitance Change:**

- o The touch draws a small amount of current, causing a **local change in capacitance**.

5. **Sensing and Processing:**

- o The sensor controller detects this change and calculates the **touch location** using the altered capacitance values from

6. **Action Triggered:**

- o The processor then performs the corresponding action (e.g., opening an app or scrolling).

Solution Scheme:

10b Describe a real-life example where a transducer is used in automation.

Solution:

Real-Life Example of a Transducer in Automation: Automatic Temperature Control in HVAC Systems.

Key Components:

1. **Transducer: Resistance Temperature Detector (RTD) or Thermistor**
 - o Converts the **temperature of the air** into an **electrical signal** (resistance change).
2. **Signal Conditioning Circuit**
 - o Converts the resistance output to a voltage or digital signal suitable for processing.
3. **Microcontroller/PLC**
 - o Reads the signal from the transducer and compares it with the **desired set temperature**.
4. **Actuator**
 - o Controls HVAC components such as **compressors, heaters, or fans** based on the controller's output.
5. **Temperature Sensing:**
 - o The **RTD sensor** detects the room temperature.
 - o Its resistance changes proportionally to the temperature.
6. **Signal Conversion:**
 - o A signal conditioner (like a Wheatstone bridge or ADC) converts this resistance into a measurable voltage.
7. **Decision Making:**
 - o The controller (like a microcontroller or PLC) compares the measured temperature with the **set point**.
 - o If the room is too cold, it activates the **heater**.
 - o If it's too hot, it turns on the **air conditioner** or **cooling fan**.
8. **Feedback Loop:**
 - o Continuous temperature monitoring maintains the desired comfort level automatically.

Solution Scheme:

Discuss the selection parameters for a transducer used in high-temperature furnace applications.

Solution:

Selection Parameters for a High-Temperature Furnace Transducer:

- Ø Temperature Range
- Ø Sensor Type
- Ø Sensor Type
- Ø Accuracy and Sensitivity
- Ø Accuracy and Sensitivity
- > Signal Output Type
- > Material Compatibility
- > Response Time
- > Mechanical Durability
- > Size and Mounting
- > Calibration and Linearity
- > Cost and Maintenance

10c

- Explanations any four

Solution Scheme: